

Revealing How Dust Grain Properties and Temperature Distributions in High-Redshift Galaxies Correlate with Star Formation Activity Using Machine Learning

Introduction

Early universe star formation rates are poorly constrained because interstellar dust absorbs UV light from young stars and re-emits it in the infrared, skewing observations. Traditional attenuation laws derived from nearby galaxies fail at $z > 5$, where dust composition and geometry evolve rapidly (Markov et al., 2023; Tress et al., 2019).

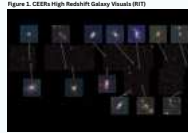


Figure 1: JWST CEERS High-Redshift Galaxy Visuals (RIT)

Prior Research

- Markov et al. (2023): detected unexpected 2175 Å UV bump at $z \sim 7-8$, showing dust composition diverges from local laws
- Fudamoto et al. (2020): found star-forming regions hidden behind dust in UV-bright high- z galaxies
- Toyouchi et al. (2025): showed dust temperature and mass evolve rapidly at $z > 5$, directly linking dust heating to star formation

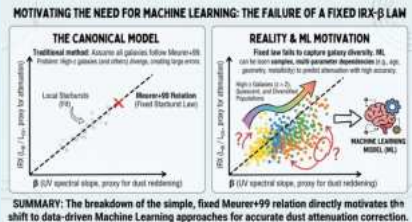
Research Objectives

Research Question: Can machine learning predict SFR in high-redshift galaxies ($z > 5$) using dust and photometric features, without assuming a fixed attenuation law?

At $z > 5$, the universe is less than 1 billion years old. Dust grain properties, temperature, and geometry are actively evolving, making locally-derived attenuation laws unreliable for SFR estimation.

Hypothesis: A Gradient Boosting model trained on UV slope β , stellar mass, E(B-V), and IRX from JWST CEERS data will predict SFR more accurately than a linear regression baseline that assumes a fixed attenuation relationship.

Figure 2: Conceptual motivation for a machine learning approach. Left: the Meurer+99 IRX- β relation assumes all galaxies follow a fixed starburst law. Right: high-redshift and diverse galaxy populations scatter significantly off this relation, motivating a data-driven model that learns attenuation without fixed law assumptions.



SUMMARY: The breakdown of the simple, fixed Meurer+99 relation directly motivates the shift to data-driven Machine Learning approaches for accurate dust attenuation correction.

Methodology

Dataset: JWST CEERS DR1 catalog (Cox et al., 2025) – 87,345 galaxies filtered to $z > 5$. Physical parameters from LePHARE SED fitting.

Figure 3: Feature completeness across CEERS DR1 sample

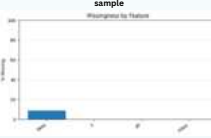


Table 1: Feature Extraction Table

Feature	Source
UV slope β	F155W / F150W fluxes
Stellar mass ($\log M_{\odot}$)	LePHARE SED fitting
E(B-V)	LePHARE SED fitting
A_{FIR}	Meurer et al. 1999
IRX proxy	Derived from A_{FIR}
Interaction terms	Computed

Models trained:

Linear Regression - fixed attenuation baseline

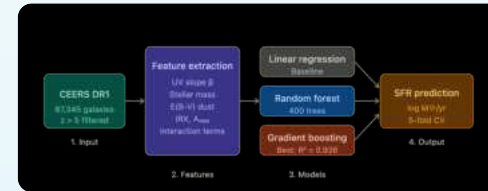
Random Forest - 400 trees

Gradient Boosting - 300 estimators, $lr = 0.05$

Validation: 80/20 train-test split + 5-fold cross-validation



Figure 4: Pipeline Flowchart



Results & Validation

Gradient Boosting outperformed the linear baseline, reducing RMSE by 14% (0.371 $\rightarrow$ 0.319 dex). 5-fold cross-validation confirmed stability across data partitions. The improvement over linear regression demonstrates that dust-SFR relationships at $z > 5$ contain non-linear structure inconsistent with fixed attenuation law assumptions.

According to the heatmap (Pearson correlation matrix), the near-perfect SFR-mass correlation ($r = 1.00$) reflects the star-forming main sequence. IRX- β collinearity ($r = 1.00$) arises from algebraic derivation and is addressed in limitations.

Figure 6: Residuals centered near zero confirm unbiased predictions across the SFR range.

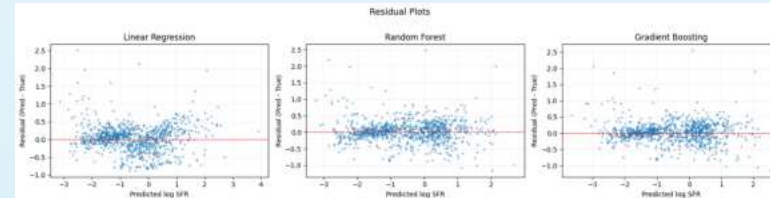


Table 2: Model Performance

Model	MAE	RMSE	R^2
Linear Regression	0.266	0.371	0.900
Random Forest (n=400)	0.225	0.333	0.920
Gradient Boosting (est. 300)	0.218	0.319	0.926

Figure 5: Predicted vs. true log SFR for all three models. R^2 values shown per panel.

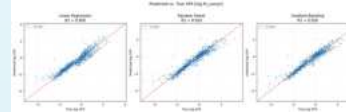
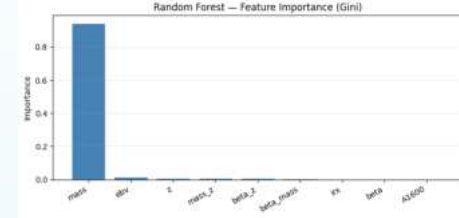


Figure 7: Random Forest Gini importance. Stellar mass dominates; E(B-V) ranks second among dust features.



Conclusion

Gradient Boosting outperformed the Linear Regression baseline ($R^2 = 0.926$ vs. 0.900), demonstrating that dust-SFR relationships in high-redshift galaxies are non-linear and cannot be captured by fixed attenuation laws. Stellar mass was the strongest predictor, consistent with the star-forming main sequence, while E(B-V) ranked second, confirming that dust attenuation carries measurable predictive signal beyond mass alone. Future work will incorporate ALMA far-infrared data for direct IRX measurements, expand to additional JWST fields including JADES and PRIMER, and explore neural network architectures for larger datasets.

Limitations

IRX derived algebraically from β creates collinearity. SFR labels from LePHARE encode mass by construction. Photometric redshift uncertainties $\sigma_{\text{N}MAD} = 0.035-0.073$ add scatter at $z > 8$.

References

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